

Multi-Robot Motion Planning for Magnetic Robots

MSc Thesis Topic

Mentors: Christoph Scherer (TU Berlin), Lara Bergmann (U Bielefeld)

Examiners: Wolfgang Hönig (TU Berlin), Klaus Neumann (U Bielefeld)

September 2026

1 Introduction

Magnetic levitation systems are commercially available off-the-shelf technologies with established applications in industrial automation. Multiple so-called tiles, each containing electromagnetic coils, can be connected to form a larger planar workspace. On this surface, so-called movers equipped with permanent magnets can travel at high speeds while carrying payloads. At the low level, the movers are actuated indirectly by controlling the currents in the electromagnetic coils. Existing controllers provide a more abstract interface based on setpoints that approximately resemble state-dependent double-integrator dynamics [3]. For example, the action space may be defined as $[a_x, a_y, a_\theta]^\top$, corresponding to the translational accelerations along the x - and y -axes and the angular acceleration around the vertical axis.

However, the feasible action space depends on the current state and position of the mover. In particular, yaw rotations beyond $\pm 10^\circ$ are only possible in specific regions of the workspace and may reduce the maximum admissible payload. Multiple movers can additionally be physically coupled to form robotic systems with an extended workspace [2] or with dedicated manipulation capabilities [1], with the latter work currently submitted to ICRA 2027.

From a motion-planning perspective, magnetic levitation systems introduce several novel algorithmic challenges. The problem constitutes a kinodynamic multi-robot planning setting for which centralized algorithms such as db-CBS [5] and db-LaCAM [4] already exist. However, in magnetic levitation systems, the available actions may be state- and location-dependent, while the physical payload capabilities can additionally depend on both the mover type and its position within the workspace. Moreover, it remains an open question how physically coupled magnetic levitation robots should best be represented during planning. Possible approaches include planning directly in the joint configuration space of the coupled system or using softer coupling formulations similar to pc-db-CBS [6].

The goal of this thesis is to extend and apply existing kinodynamic multi-robot motion-planning algorithms to the specific characteristics of magnetic movers and magnetic robotic systems. The resulting methods should be evaluated systematically by benchmarking different algorithms, or suitable adaptations thereof, with respect to success rate, solution quality, and computational effort. Finally, the generated trajectories are expected to be validated experimentally on a real magnetic levitation system at Bielefeld University.



Figure 1: 6D-platform MagBot physically coupling two magnetically levitating movers into a single robotic system.

2 Milestones

1. Literature survey: kinodynamic multi-robot motion planning, magnetic levitation, magnetic robotics
2. Benchmark: definition of expressive benchmark problems that require planning using state dependent action spaces
3. Single mover case: develop and compare planners for a single mover that can handle the state-dependent action space. Suggested options: kinodynamic RRT and an extension of db-A*.
4. Multiple mover case: develop one planner that can plan for multiple ($n \geq 8$) movers. Suggested option: extension of db-ECBS.
5. MagBot case: compare joint-space planning of two movers (e.g., via kinodynamic RRT or db-A*) with a dedicated planner for physically coupled robots (such as pc-db-CBS).
6. Deployment on a real system (in Bielefeld!)

3 Prerequisites

1. Experience with motion planning, ideally for mobile robots
2. Strong experience with C++
3. Background in nonlinear optimization

4 Bibliography

- [1] L. Bergmann, N. Greis, C. Grothues, L.-M. Weigelt, and K. Neumann. “From Transportation to Manipulation: Enabling Grasping in Magnetic Robotics”. 2026. arXiv: [2609.12883 \[cs.R0\]](#).
- [2] L. Bergmann, N. Greis, and K. Neumann. “From Transportation to Manipulation: Transforming Magnetic Levitation to Magnetic Robotics”. Version 1. 2026. Pre-published.
- [3] L. Bergmann, C. Grothues, and K. Neumann. “MagBotSim: Physics-Based Simulation and Reinforcement Learning Environments for Magnetic Robotics”. Version 1. 2025. Pre-published.
- [4] A. Moldagalieva, K. Okumura, A. Prorok, and W. Hönig. “db-LaCAM: Fast and Scalable Multi-Robot Kinodynamic Motion Planning with Discontinuity-Bounded Search and Lightweight MAPF”. 2025. arXiv: [2512.06796](#).
- [5] A. Moldagalieva, J. Ortiz-Haro, M. Toussaint, and W. Hönig. “db-CBS: Discontinuity-Bounded Conflict-Based Search for Multi-Robot Kinodynamic Motion Planning”. In: *2024 IEEE International Conference on Robotics and Automation (ICRA)*. 2024, pp. 14569–14575.
- [6] K. Wahba and W. Hönig. “pc-dbCBS: Kinodynamic Motion Planning of Physically-Coupled Robot Teams”. In: *IEEE Robotics and Automation Letters (R-AL)* (2025).